

Python Version Python 3.8 or higher is required.

Hardware A CUDA-compatible GPU is recommended. CPU execution is supported but will result in significantly longer training times (estimated 12–24 hours for full 30-run experiment vs 2–4 hours on GPU).

Dependencies Install all required libraries using the following command:

```
“pip install torch transformers scikit-learn pandas numpy matplotlib scipy nltk”
```

Exact versions used in this study:

```
torch==2.0.1
```

```
transformers==4.35.0
```

```
scikit-learn==1.3.0
```

```
pandas==2.0.3
```

```
numpy==1.24.3
```

```
matplotlib==3.7.2
```

```
scipy==1.11.1
```

```
nltk==3.8.1
```

NLTK Data The following NLTK corpus is required and will be downloaded automatically on first run:

```
stopwords
```

If automatic download fails, run the following manually:

```
import nltk
```

```
nltk.download('stopwords')
```